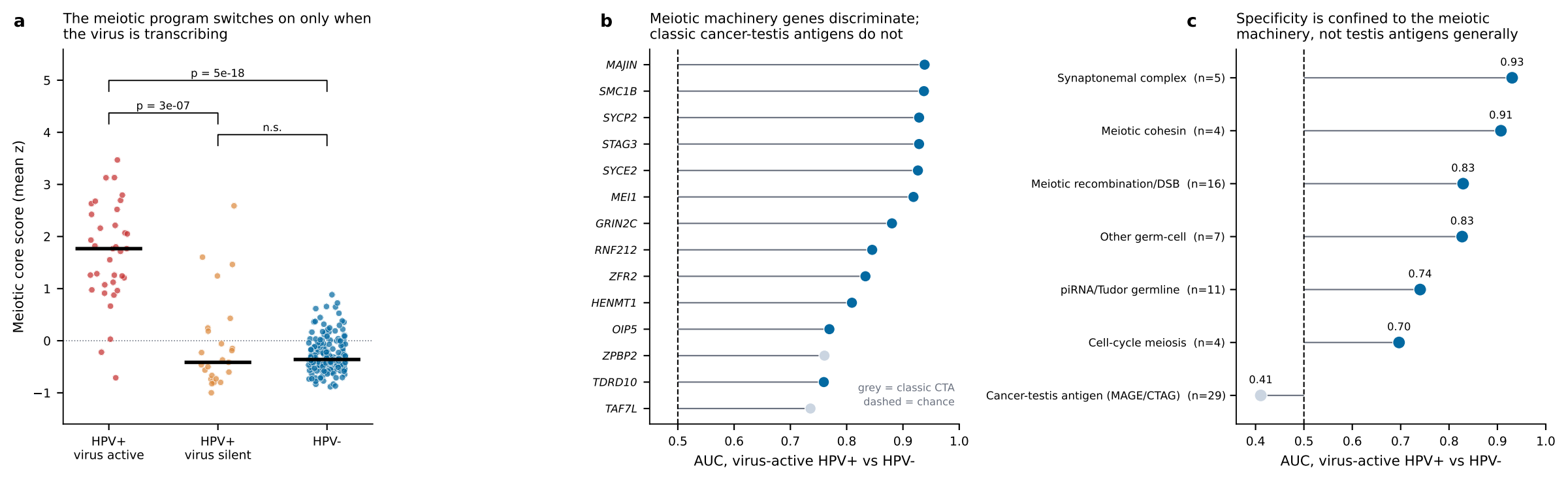
a
The meiotic program switches on only when the virus is transcribing
p = 5e-18
p = 3e-07
n.s.
Meiotic core score (mean z)
5
4
3
2
1
0
−1
HPV+ virus active
HPV+ virus silent
HPV-
b
Meiotic machinery genes discriminate; classic cancer-testis antigens do not
MAJIN
SMC1B
SYCP2
STAG3
SYCE2
MEI1
GRIN2C
RNF212
ZFR2
HENMT1
OIP5
ZPBP2
TDRD10
TAF7L
grey = classic CTA
dashed = chance
0.5
0.6
0.7
0.8
0.9
1.0
AUC, virus-active HPV+ vs HPV-
c
Specificity is confined to the meiotic machinery, not testis antigens generally
Synaptonemal complex (n=5)
0.93
Meiotic cohesin (n=4)
0.91
Meiotic recombination/DSB (n=16)
0.83
Other germ-cell (n=7)
0.83
piRNA/Tudor germline (n=11)
0.74
Cell-cycle meiosis (n=4)
0.70
Cancer-testis antigen (MAGE/CTAG) (n=29)
0.41
0.4
0.5
0.6
0.7
0.8
0.9
1.0
AUC, virus-active HPV+ vs HPV-